ARPITA MALLIK

+880-1552398872 | arpitamallik13@gmail.com | Portfolio - Arpita Mallik

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Chittagong - Bangladesh

Summary

I am a final-year Computer Science & Engineering student with a strong interest in Machine Learning, NLP, and Large Language Models (LLMs). I'm committed to continuous learning and growth, both academically and professionally. My involvement in research and technical activities has sharpened my problem-solving skills and analytical thinking. I'm eager to grow further and make a positive impact through my work in the field of technology.

Experience

- **Green Lead** *June 2024 - Present*
  Tech Solutions Architect Remote
  - Maintain and update the Green Lead WordPress website, ensuring optimal performance and user experience.
  - Assist in troubleshooting and resolving technical issues related to the website.

Education

- **Chittagong University of Engineering & Technology** *March 2022 - Present*
  BSc. in Computer Science & Engineering
  - GPA: 3.59/4.00 (Till 7th semester)

Skills

- **Programming Languages:** Python, C, C++
- **Database Systems:** MySQL
- **Frameworks & Libraries:** PyTorch, FastAPI, Flask, Streamlit
- **Tools:** Git, GitHub, Docker, MLflow, Power BI, LaTeX

Projects

- **Vehicle Insurance Prediction – End-to-End MLOps System**
  Tools: Python, scikit-learn, MongoDB Atlas, AWS (S3, ECR, EC2, IAM), Docker, GitHub Actions, Flask
  - Built a modular end-to-end ML pipeline for vehicle insurance prediction.
  - Implemented schema-based validation and artifact-driven model training and evaluation.
  - Designed a versioned S3 model registry with threshold-based promotion.
  - Deployed a Dockerized Flask service on AWS EC2 with CI/CD via GitHub Actions.
- **Medical AI Chatbot – Memory-Based RAG Medical QA System**
  Tools: Python, Flask, LangChain, Pinecone, HuggingFace Embeddings
  - Developed a medical QA chatbot using a memory-augmented RAG pipeline with semantic retrieval via Pinecone.
  - Implemented Flask backend with LangChain for chunking, retrieval, and session-aware conversational history.
- **Movie Recommendation System**
  Tools: Python, Streamlit
  - Built a recommendation system using ML algorithms to suggest movies based on user preferences.
  - Deployed the model with Streamlit to provide an interactive and lightweight user interface.
- **Cyberbullying Detection App**
  Tools: Python, Flask, HTML5, CSS
  - Developed a web app to identify toxic and bullying comments using NLP techniques.
  - Trained classification models and integrated the prediction system with a Flask-based frontend.
- **End-to-End Boston House Price Prediction**
  Tools: Python, Flask, Docker, Git, GitHub
  - Built a full-stack ML web app to predict house prices using features like crime rate and tax.
  - Dockerized the app for seamless deployment and added REST API support with Postman.
- **Medical Insurance Charges Prediction**
  Tools: Python, FastAPI, XGBoost, HTML, JavaScript
  - Predicted medical insurance charges using an ML model based on user health and demographics.
  - Built a FastAPI backend and web UI to allow real-time predictions and input validation.
- **PowerBI Online Courses Analysis Dashboard**
  Tools: Power BI
  - Created a Power BI dashboard to explore trends in online learning platforms.
  - Visualized course types, skill demand, ratings, and language preferences for decision-making.

- **CUET Thesis & Project Portal**
  Tools: React JS, Firebase
  - Built a centralized portal for managing and browsing CUET students' thesis and projects.
  - Used Firebase for real-time storage, retrieval, and user authentication.
- **TuitionEase**
  Tools: HTML, CSS, JavaScript, PHP, MySQL
  - Developed a web platform for tuition management including tutor schedules and student profiles.
  - Designed both frontend and backend for seamless navigation and conflict-free scheduling.
- **ZenSpace – Meditation Based Software**
  Tools: Javaswing, MySQL
  - Created a Java desktop app for guided meditation, journaling, and donation processing.
  - Designed a clean UI with Java Swing and stored session data securely in MySQL.
- **Green Lead Official Website**
  Tools: WordPress
  - Designed and maintained a WordPress website to raise awareness for climate action.
  - Managed content and optimized performance for public engagement and clarity.

Publications

C=Conference, J=Journal,S=In Submission, T=Thesis

[C.1] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*
DOI: 10.18653/v1/2025.dravidianlangtech-1.57

[C.2] **CUET-823 at MAHED 2025 Shared Task: Large Language Model-Based Framework for Emotion, Offensive, and Hate Detection in Arabic**. Ratnajit Dhar, Arpita Mallik. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, Suzhou, China, November 2025, pp. 620–625.
DOI: 10.18653/v1/2025.arabicnlp-sharedtasks.84

Honors and Awards

Finalist – DL Sprint 4.0 *February 2026*
BUET CSE Fest 2026
- Ranked among the Top 18 teams out of 192 in a nationwide deep learning competition.
- Built a pipeline for Bangla long-form speech recognition and speaker diarization using Whisper-based ASR and fine-tuned diarization models.

Champion – NodeScape *July 2025*
CUET Computer Club
- Secured 1st place in CUET Computer Club's flagship 3-phase hackathon, NodeScape.
- Developed an AI-powered graph classification system with algorithm visualization and Dockerized CI/CD deployment using React, Flask, and Python.

Finalist – Intra CUET Machine Learning Contest *July 2025*
IEEE CS CUET Student Branch Chapter
- Selected as a finalist in intra-university ML contest.
- Built a Bangla physics MCQ question-answering system leveraging large language models (LLMs) to understand and solve concept-based queries.

Top 7 Finalist – HackCSB 2024 *August 2024*
Coding Shikhbe Bangladesh
- Achieved finalist position out of 116 teams at HackCSB for the project "MotivationMate."
- Demonstrated creativity and teamwork in building a mental health-focused web application.

Certifications

- **Google Cloud Fundamentals: Core Infrastructure** - Coursera *June 2025*
- **Career Essentials in Generative AI** - Microsoft & LinkedIn *July 2024*
- **Supervised Machine Learning: Regression and Classification** - Coursera *February 2024*
- **Advanced Learning Algorithms** - Coursera

Leadership Experience

- **Vice Chair (Machine Learning Wing)** *July 2025 – Present*
  IEEE CS CUET Student Branch Chapter